Exercise Sheet 27

All modules are finite-dimensional over $\mathbb{C}$. "Reciprocity" and the "isotypic commutant formula" refer to Lecture 27. Exercises 27.1–27.3 and 27.6 belong to the first part of Lecture 27, on invariant tensors and contractions. Exercises 27.4 and 27.5 belong to its second part, on generation and representation rings.

Exercise 27.1 (Invariants for $\mathfrak{sl}_2$). Let $V_1 = \mathbb{C}^2$ with basis e_1, e_2, and let $\beta(u,v) = \det(u\ v)$.

(i) Decompose $V_1^{\otimes 3}$ and $V_1^{\otimes 4}$. Find $\dim \mathrm{End}_{\mathfrak{sl}_2}(V_1^{\otimes 2})$ and $\dim \mathrm{End}_{\mathfrak{sl}_2}(V_1^{\otimes 3})$.

(ii) Deduce $\dim (V_1^{\otimes 4})^{\mathfrak{sl}_2}$ and $\dim (V_1^{\otimes 6})^{\mathfrak{sl}_2}$ using duality and the identification of intertwiners with invariants.

(iii) Let $\Omega = e_1 \otimes e_2 - e_2 \otimes e_1$ and $E(u \otimes v) = \beta(u,v)\Omega$. Let $P(u \otimes v) = v \otimes u$. Prove $1 - P = E$ on $V_1 \otimes V_1$. Why does this not contradict the three matchings of four points?

Exercise 27.2 (Degree-three invariants for $\mathfrak{sl}_3$). Let $V = \mathbb{C}^3$ and $\mathfrak{g} = \mathfrak{sl}_3$.

(i) Prove $\dim (V^{\otimes 3})^{\mathfrak{g}} = 1$, and show that vol spans it. Is vol invariant under $\mathfrak{gl}_3$?

(ii) Prove $(V \otimes V \otimes V^*)^{\mathfrak{g}} = 0$.

(iii) This part returns to Exercise 19.6 with the tools of Lecture 27. Using reciprocity, prove that the invariant trilinear forms on $\mathfrak{g}$ form a two-dimensional space with basis

$$d(Y,Z,W) = \mathrm{tr}\big(Y(ZW + WZ)\big), \qquad f(Y,Z,W) = \mathrm{tr}\big(Y[Z,W]\big).$$

Exercise 27.3 (Volume tensors).

(i) Prove the two formulas stated in Lecture 27: show that $g^{\otimes N}\mathrm{vol} = \det(g)\,\mathrm{vol}$ for $g \in GL_N$, and that $X \in \mathfrak{gl}_N$ acts on vol by $\mathrm{tr}\, X$.

(ii) For $\mathfrak{so}_3$ on $V = \mathbb{C}^3$, prove $(V^{\otimes 3})^{\mathfrak{so}_3} = \mathbb{C}\,\mathrm{vol}$ and $(V^{\otimes 3})^{O_3} = 0$. Identify the matching intertwiner $V \otimes V \to V$.

(iii) Let $\omega = \sum_i e_i \wedge f_i \in \Lambda^2 \mathbb{C}^{2n}$. Prove $\omega^{\wedge n} = n!\, e_1 \wedge f_1 \wedge \cdots \wedge e_n \wedge f_n$.

(iv) For $N = 4$, prove $\dim \mathrm{End}_{\mathfrak{so}_4}(V \otimes V) = 4$. Prove that the elements that also commute with $r^{\otimes 2}$, where $r = \mathrm{diag}(-1,1,1,1)$, are the span of $1, P, E$.

Exercise 27.4 (Faithful but not tensor-generating). A module V *tensor-generates* if every simple module is a summand of some $V^{\otimes k} \otimes V^{*\otimes l}$.

(i) Prove that every weight of every such summand lies in the subgroup $\Lambda_V \subseteq P$ generated by the weights of V.

(ii) Show that the adjoint $\mathfrak{sl}_2$-module V_2 is faithful but does not tensor-generate. In $K(\mathfrak{sl}_2) = \mathbb{Z}[x]$, show that the subring generated by $[V_2]$ does not contain $[V_1]$.

(iii) Show the same for the vector module of $\mathfrak{so}_N$, $N \geq 5$.

Exercise 27.5 (The representation ring of $\mathfrak{sl}_3$). Let $V = \mathbb{C}^3$ be the defining module. Let $x = [L(\omega_1)] = [V]$ and $y = [L(\omega_2)]$ in $K(\mathfrak{sl}_3)$.

(i) Express $[L(2\omega_1)]$, $[L(\omega_1+\omega_2)]$ and $[L(3\omega_1)]$ as polynomials in x, y. Check the dimensions.

(ii) Decompose x^3, and compare it with Schur–Weyl duality. What is $\dim \mathrm{End}_{\mathfrak{sl}_3}(V^{\otimes 3})$?

Exercise 27.6 (Orthogonal and symplectic contractions). Use the operators $1, P, E$ on $V \otimes V$. In the symplectic case set $E(u \otimes v) = \beta(u, v)\Omega_{\mathrm{sp}}$; in the orthogonal case use the definition in Lecture 27.

(i) For $\mathfrak{sp}_{2n}$, prove $E^2 = 2nE$ and $PE = EP = -E$. Figure 27.2 draws the first two of these relations. Verify that $p_0 = \frac{1}{2n}E$, $p_S = \frac{1}{2}(1+P)$ and $p_A = \frac{1}{2}(1-P) - \frac{1}{2n}E$ are orthogonal idempotents with sum 1.

(ii) Compute their ranks from traces. Check the case of $\mathfrak{sp}_4$.

(iii) Repeat the trace computation for $\mathfrak{so}_N$, returning to the projectors of the worked example in Lecture 27. Explain the case $n = 1$ in (ii).